UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Problems on Taylor's and Maclaurin's Series

Complete study pack for class 008: Taylor expansions of functions of two variables about a point, Maclaurin expansions about the origin, and approximate values from a Taylor polynomial

Compiled from the PESU class 008 slide deck and AV-summary transcript, the Unit 1 class-work sheet and past ESA papers. Generated 14 October 2026.

· How to use this pack

Topic 1 expands about a point (a, b), topic 2 about the origin, topic 3 uses the expansion to approximate a value. Every problem follows the same four steps (Figure F1.1). The theory (why series, one-variable series, the statements) is in the class 007 pack "Taylor's and Maclaurin's Expansion"; the formulas are repeated here so that every problem can be solved from this pack alone. The past-paper and class-work problems also appear in that pack; the answers agree. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b

Contents

1. 1 · Taylor expansion about a point (a, b)
2. 2 · Maclaurin expansion about the origin
3. 3 · Approximate values from a Taylor polynomial
4. Appendix A · Symbol glossary
5. Appendix B · Sources, coverage and gaps

1 Taylor expansion about a point (a, b)

Write $h = x - a$, $k = y - b$, tabulate f and its partial derivatives, evaluate them at (a, b) and substitute into $f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \frac{1}{3!}\{\dots\}$. "In powers of $(x-1)$ and $(y - \frac{\pi}{4})$" means about $(1, \frac{\pi}{4})$. Class 008 works $e^x \cos y$, $\sin(xy)$ and $xy^2 + y\cos(x-y)$; the ESA favourite is $\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$.

Key terms and definitions

Expansion "in powers of (x − a) and (y − b)". Taylor's series of $f(x, y)$ about the point (a, b).
S008·3 · T008 00:00:37

Up to n-th degree terms. Keep every term whose total power of $(x-a)$ and $(y-b)$ is at most n.
ADDED FOR CLARITY T008 00:06:10

> "$f(x,y) \approx f(a,b) + (x-a)f_x(a,b) + (y-b)f_y(a,b) + \frac{1}{2!}\big[(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b)f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)\big]$" S008·14 · S008·19

Formulas

E1.1 · Taylor's series about (a, b), to second order ESA ★★★

$$f(x,y) = f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\} + \dots, \quad h = x - a,\ k = y - b$$

GIVES	The polynomial to substitute into.
APPLIES	All derivatives evaluated at (a, b).
NOTE	h, k are shorthand added for clarity.

S008·3 · S008·14 · T008 00:00:37 · T008 00:01:25

E1.2 · Third-degree term ESA ★★★

$$+\frac{1}{3!}\{h^3 f_{xxx} + 3h^2 k f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\}$$

GIVES	The terms for "up to third degree".
APPLIES	Add to E1.1 or E2.1.
NOTE	Coefficients 1, 3, 3, 1; asked in four of six ESA questions.

S008·10 · ESA Dec-23 1b · ESA Jun-24 1c

Graphs and diagrams

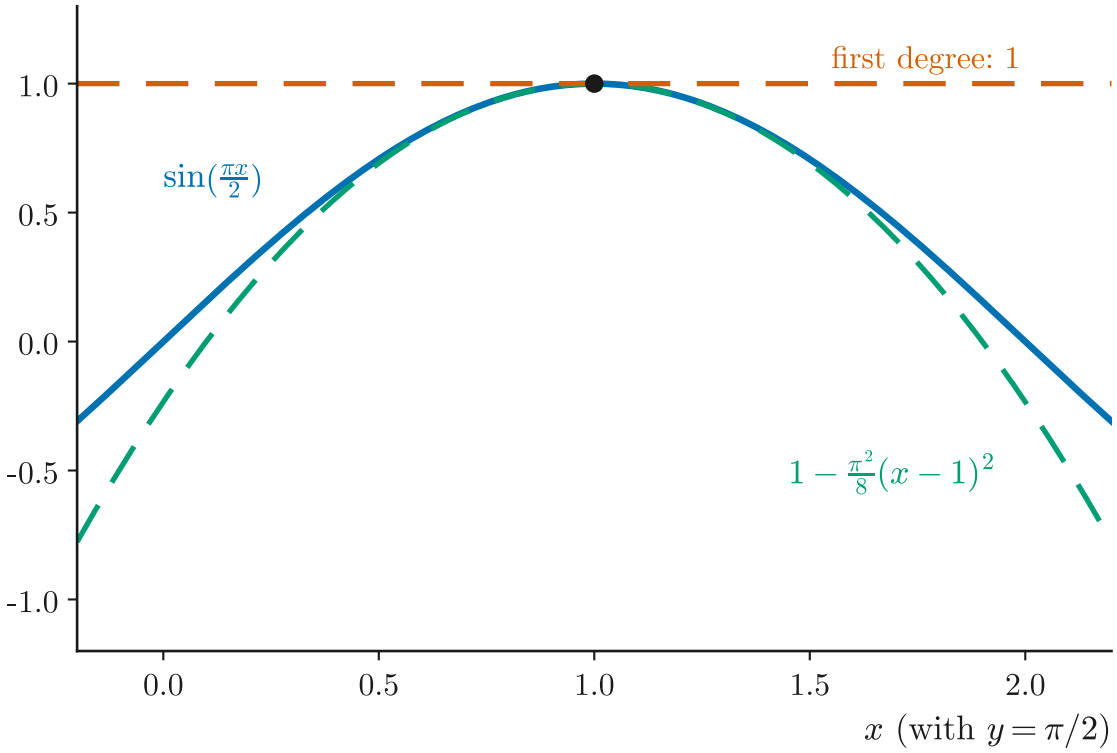
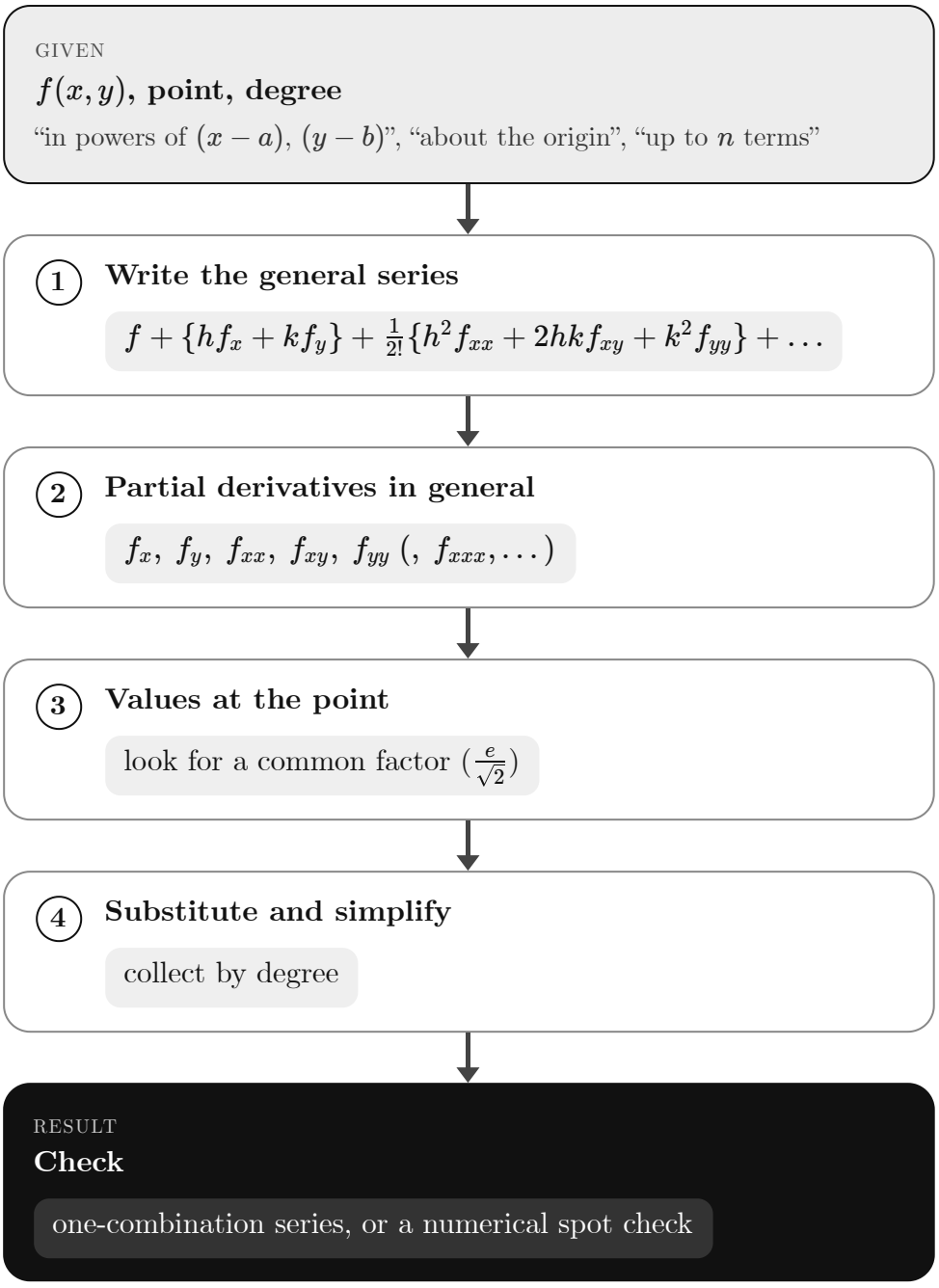


Figure G1.1. $\sin(xy)$ near $(1, \pi/2)$: a slice at $y = \pi/2$. ESA ★

ADDED FOR CLARITY S008·14 · S008·16

How to read it. Along $y = \frac{\pi}{2}$, $\sin(xy) = \sin\frac{\pi x}{2}$ (blue) has a maximum at $x = 1$, so both first partials vanish and the first-degree polynomial is the flat line 1 (vermilion). The second-degree polynomial $1 - \frac{\pi^2}{8}(x-1)^2$ (green) follows the curvature near $x = 1$.

Flowcharts



Remark. For "up to three terms", stop at three non-zero terms; for "third degree", include the $\frac{1}{3!}$ bracket.

Figure F1.1. Solving a Taylor or Maclaurin expansion problem ESA ★★★

T008 00:00:21 · T008 00:02:07 · T008 00:05:19

Important points and why

1. Write the general series first, then find the derivatives, then substitute. That is the order the lecture follows. ESA ★★ T008 00:00:21 · T008 00:02:07

2. For $e^x \cos y$ at $(1, \frac{\pi}{4})$ every derivative is $\pm\frac{e}{\sqrt{2}}$; only the signs differ, so take $\frac{e}{\sqrt{2}}$ out as a common factor. ESA ★★★ T008 00:04:54 · T008 00:05:19
3. If the question asks for third degree, you need the third-order partials $f_{xxx}, f_{xxy}, f_{xyy}, f_{yyy}$ too. ESA ★★★ T008 00:06:10
4. Derivatives that vanish at the point drop whole terms: for $\sin(xy)$ at $(1, \frac{\pi}{2})$ both first partials are 0. ESA ★★ S008·15 · S008·16
5. Differentiate carefully with the product and chain rules; the slide's own f_{yy} for $xy^2 + y\cos(x-y)$ has a sign slip (see the exam traps). ESA ★★ ADDED FOR CLARITY S008·17 · S008·18

Exam trap. For $xy^2 + y\cos(x-y)$: $f_{yy} = 2x + 2\sin(x-y) - y\cos(x-y)$, so $f_{yy}(1,1) = 1$, not 3 as slide 008·18 prints. ADDED FOR CLARITY S008·17 · S008·18

Exam trap. Every value on slide 008·3 is $\pm\frac{e}{\sqrt{2}}$, not $\pm\frac{e}{2}$ as said at T008 00:03:47.
ADDED FOR CLARITY S008·3 · T008 00:03:47

Worked examples

W1.1 · $e^x \cos y$ about $(1, \frac{\pi}{4})$ (slides 008·3–4) ESA ★★★

GIVEN Expand $e^x \cos y$ in powers of $(x-1)$ and $\left(y - \frac{\pi}{4}\right)$ up to second degree. Let $h = x - 1$, $k = y - \frac{\pi}{4}$.

STEP 1 $f = e^x \cos y = \frac{e}{\sqrt{2}}$; $f_x = e^x \cos y = \frac{e}{\sqrt{2}}$; $f_y = -e^x \sin y = -\frac{e}{\sqrt{2}}$.

STEP 2 $f_{xx} = e^x \cos y = \frac{e}{\sqrt{2}}$; $f_{xy} = -e^x \sin y = -\frac{e}{\sqrt{2}}$; $f_{yy} = -e^x \cos y = -\frac{e}{\sqrt{2}}$.

STEP 3 E1.1, taking $\frac{e}{\sqrt{2}}$ out: $\frac{e}{\sqrt{2}}\left[1 + h - k + \frac{1}{2!}(h^2 - 2hk - k^2)\right]$.

ANSWER
$$e^x \cos y \approx \frac{e}{\sqrt{2}}\left[1 + (x-1) - \left(y - \tfrac{\pi}{4}\right) + \tfrac{1}{2}\left\{(x-1)^2 - 2(x-1)\left(y - \tfrac{\pi}{4}\right) - \left(y - \tfrac{\pi}{4}\right)^2\right\}\right]$$

CHECK sympy series agrees. Same question as ESA May 2024 Q1.b.

S008·3 · S008·4 · T008 00:00:21 · T008 00:05:53 · ESA May-24 1b

W1.2 · $\sin(xy)$ about $(1, \frac{\pi}{2})$ (slides 008·14–16) ESA ★★

GIVEN Expand $\sin(xy)$ in powers of $(x-1)$ and $\left(y-\frac{\pi}{2}\right)$ up to second-degree terms. Let $h = x-1$, $k = y-\frac{\pi}{2}$.

STEP 1 $f = \sin\frac{\pi}{2} = 1$; $f_x = y\cos(xy) = 0$; $f_y = x\cos(xy) = 0$.

STEP 2 $f_{xx} = -y^2\sin(xy) = -\frac{\pi^2}{4}$; $f_{xy} = \cos(xy) - xy\sin(xy) = -\frac{\pi}{2}$; $f_{yy} = -x^2\sin(xy) = -1$.

STEP 3 E1.1: $1 + \frac{1}{2}\big[-\frac{\pi^2}{4}h^2 - \pi hk - k^2\big]$.

ANSWER $\boxed{\sin(xy) \approx 1 - \frac{\pi^2}{8}(x-1)^2 - \frac{\pi}{2}(x-1)\left(y-\tfrac{\pi}{2}\right) - \frac{1}{2}\left(y-\tfrac{\pi}{2}\right)^2}$

CHECK sympy series agrees; Figure G1.1.

S008·14 · S008·15 · S008·16

W1.3 · $xy^2 + y\cos(x-y)$ about $(1,1)$ (slides 008·17–19) ESA ★★

GIVEN Find the Taylor series of $f = xy^2 + y\cos(x-y)$ up to second order about $(1,1)$. Let $h = x-1$, $k = y-1$.

STEP 1 $f_x = y^2 - y\sin(x-y)$, $f_y = 2xy + y\sin(x-y) + \cos(x-y)$.

STEP 2 $f_{xx} = -y\cos(x-y)$, $f_{xy} = 2y + y\cos(x-y) - \sin(x-y)$, $f_{yy} = 2x + 2\sin(x-y) - y\cos(x-y)$.

STEP 3 At $(1,1)$: $f = 2$, $f_x = 1$, $f_y = 3$, $f_{xx} = -1$, $f_{xy} = 3$, $f_{yy} = 1$ (not 3, as slide 008·18 prints).

STEP 4 E1.1: $2 + (h+3k) + \frac{1}{2}(-h^2 + 6hk + k^2)$.

ANSWER $\boxed{2 + (x-1) + 3(y-1) + \frac{1}{2}\big[-(x-1)^2 + 6(x-1)(y-1) + (y-1)^2\big]}$

CHECK Slide 008·19 and the CW answer agree; sympy agrees.

S008·17 · S008·18 · S008·19 · CW U1 P28

W1.4 · sin x sin y about (π/4, π/4), second order (ESA) ESA ★★★

GIVEN Expand $\sin x \sin y$ up to second-order terms about $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$. Let $h = x - \frac{\pi}{4}$, $k = y - \frac{\pi}{4}$.

STEP 1 $f = f_x = f_y = \frac{1}{2}$.

STEP 2 $f_{xx} = f_{yy} = -\sin x \sin y = -\frac{1}{2}$; $f_{xy} = \cos x \cos y = \frac{1}{2}$.

STEP 3 $\frac{1}{2} + \frac{1}{2}(h+k) + \frac{1}{2}\{-\frac{1}{2}h^2 + hk - \frac{1}{2}k^2\}$.

ANSWER $\boxed{\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2}$

CHECK CW answer agrees; sympy agrees.

ESA Jul-23 1b · CW U1 P24

W1.5 · sin x sin y, third degree (ESA) ESA ★★★

GIVEN The same function up to third-degree terms.

STEP 1 $f_{xxx} = f_{xxy} = f_{xyy} = f_{yyy} = -\frac{1}{2}$ at $(\frac{\pi}{4}, \frac{\pi}{4})$.

STEP 2 E1.2: $\frac{1}{6} \cdot \big(-\frac{1}{2}\big)(h^3 + 3h^2k + 3hk^2 + k^3) = -\frac{1}{12}(h+k)^3$.

ANSWER $\boxed{\frac{1}{2} + \frac{1}{2}(h+k) - \frac{1}{4}(h-k)^2 - \frac{1}{12}(h+k)^3}$

CHECK sympy series to third order agrees.

ESA Dec-23 1b

W1.6 · $\sqrt{1+x+y^2}$ in powers of $(x-1)$ and y (CW) ESA ★★

GIVEN Expand $\sqrt{1+x+y^2}$ about $(1,0)$ up to second degree. Let $h = x-1$, $k = y$.

STEP 1 $f = \sqrt{2}$, $f_x = \frac{\sqrt{2}}{4}$, $f_y = 0$.

STEP 2 $f_{xx} = -\frac{\sqrt{2}}{16}$, $f_{xy} = 0$, $f_{yy} = \frac{\sqrt{2}}{2}$.

STEP 3 $\sqrt{2} + \frac{\sqrt{2}}{4}h + \frac{1}{2}\big(-\frac{\sqrt{2}}{16}h^2 + \frac{\sqrt{2}}{2}k^2\big)$.

ANSWER $\boxed{\sqrt{2}\Big[1 + \dfrac{x-1}{4} - \dfrac{(x-1)^2}{32} + \dfrac{y^2}{4}\Big]}$

CHECK CW answer agrees; sympy agrees.

CW U1 P27

Past-paper questions

ESA May 2024 (7 marks) UE22MA141A Q1.b · solved in W1.1

Expand $f(x,y) = e^x \cos y$ in powers of $(x-1)$ and $\big(y - \frac{\pi}{4}\big)$ using Taylor's series up to second degree terms.

ESA Jul 2023 (7 marks) UE22MA141A Q1.b · solved in W1.4

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to second order terms about the point $\big(\frac{\pi}{4}, \frac{\pi}{4}\big)$.

ESA Dec 2023 (7 marks) UE22MA141A Q1.b · solved in W1.5

Expand $f(x,y) = \sin x \sin y$ in Taylor's series up to third degree term about $\big(\frac{\pi}{4}, \frac{\pi}{4}\big)$.

SHORT-NOTE RECAP

About (a,b): $h = x-a$, $k = y-b$; table of $f, f_x, f_y, f_{xx}, f_{xy}, f_{yy}$ at the point; $f + \{hf_x + kf_y\} + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\}$. $e^x \cos y$ at $(1, \frac{\pi}{4})$: $\frac{e}{\sqrt{2}}[1 + h - k + \frac{1}{2}(h^2 - 2hk - k^2)]$. $\sin(xy)$ at $(1, \frac{\pi}{2})$: $1 - \frac{\pi^2}{8}h^2 - \frac{\pi}{2}hk - \frac{1}{2}k^2$. $xy^2 + y\cos(x-y)$ at $(1,1)$: $2 + h + 3k + \frac{1}{2}(-h^2 + 6hk + k^2)$.

2 Maclaurin expansion about the origin

The same method with $(a, b) = (0, 0)$, so $h = x$, $k = y$. Class 008 works $\log(2x + y + 1)$, e^{x-y} and $e^x \log(1 + y)$; the ESA papers ask $\log(1 + x - y)$, $e^{ax} \sin by$ and $\sin(x + 2y)$, all up to third degree. When f depends on one combination such as $2x + y$ or $x - y$, the one-variable series gives the answer at once and is the best check.

Key terms and definitions

Up to three terms. Keep going until there are three non-zero terms: "make sure that you will get three non-zero terms". S008·5 · T008 00:07:38

> *"Taylor's series about* $(0, 0)$ *is given by* $f(x, y) \approx f(0, 0) + f_x(0, 0)\, x + f_y(0, 0)\, y + \frac{1}{2!}\left[f_{xx}(0, 0)x^2 + 2f_{xy}(0, 0)xy + f_{yy}(0, 0)y^2\right] + \frac{1}{3!}\left[f_{xxx}(0, 0)x^3 + 3f_{xxy}(0, 0)x^2y + 3f_{xyy}(0, 0)xy^2 + f_{yyy}(0, 0)y^3\right] + \ldots$" S008·10

Formulas

E2.1 · Maclaurin's series (about the origin) ESA ★★★

$$f(x, y) = f(0, 0) + \{xf_x + yf_y\} + \frac{1}{2!}\{x^2 f_{xx} + 2xyf_{xy} + y^2 f_{yy}\} + \ldots$$

gives	The polynomial about $(0, 0)$.
applies	Derivatives at $(0, 0)$.
note	Slide 008·10 calls this "Taylor's series about (0,0)"; it is the same thing.

S008·10 · S008·12 · T008 00:06:58

E2.2 · One-combination check ESA ★★

$$f = g(t),\ t = px + qy \implies f(x, y) = g(0) + g'(0)\, t + \frac{g''(0)}{2!}t^2 + \frac{g'''(0)}{3!}t^3 + \ldots$$

gives	The same expansion in one line.
applies	f depends on x and y only through $px + qy$.
note	$\log(2x + y + 1) = t - \frac{t^2}{2} + \frac{t^3}{3}$ with $t = 2x + y$; $e^{x-y} = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}$ with $t = x - y$.

ADDED FOR CLARITY S008·6 · S008·11

Graphs and diagrams

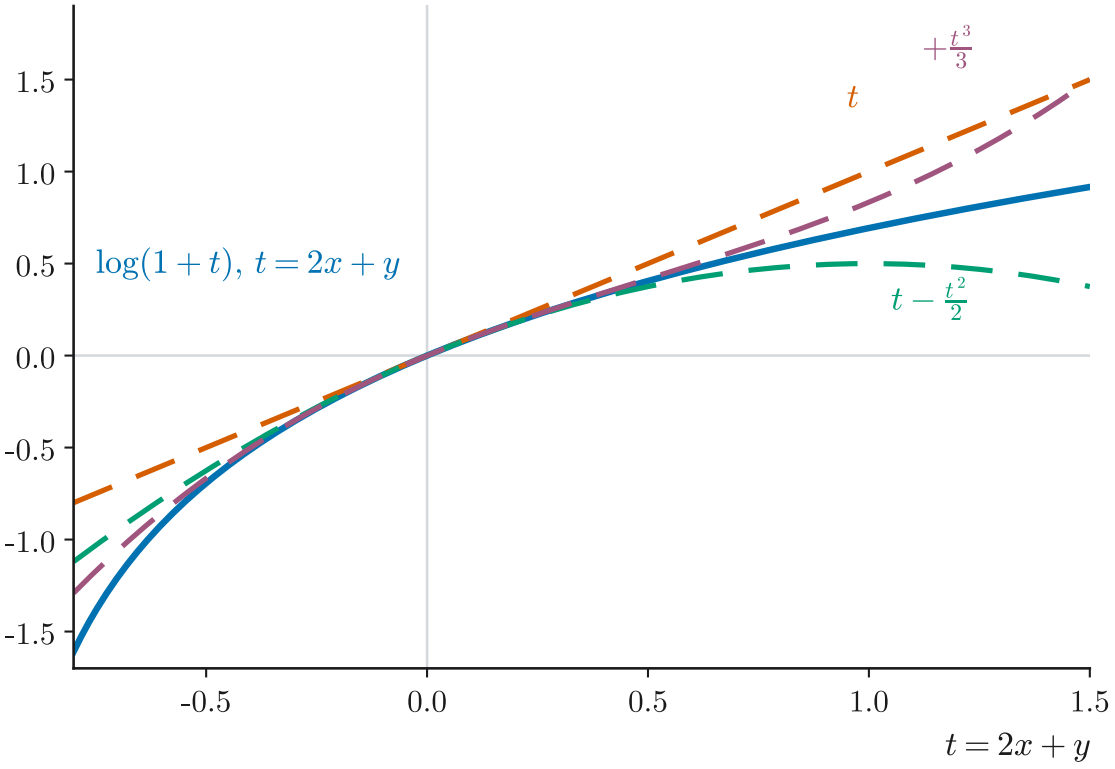


Figure G2.1. $\log(2x+y+1)$ as a function of $t=2x+y$. ESA ★★

ADDED FOR CLARITY S008·5 · S008·6

How to read it. The function depends only on $t=2x+y$, so its Maclaurin polynomials are t, $t-\frac{t^2}{2}$ and $t-\frac{t^2}{2}+\frac{t^3}{3}$. Each extra degree follows $\log(1+t)$ (blue) further from $t=0$; all fail as $t\to -1$, where the logarithm blows up.

Quick checks with a one-variable series

Function	Combination t	One-variable series	Problem
$\log(2x+y+1)$	$2x+y$	$t-\frac{t^2}{2}+\frac{t^3}{3}-\dots$	W2.1
e^{x-y}	$x-y$	$1+t+\frac{t^2}{2}+\frac{t^3}{6}+\dots$	W2.2
$\log(1+x-y)$	$x-y$	$t-\frac{t^2}{2}+\frac{t^3}{3}-\dots$	W2.4
$\sin(x+2y)$	$x+2y$	$t-\frac{t^3}{6}+\dots$	W2.6
$e^x\log(1+y)$, $e^{ax}\sin by$	product	multiply the two series, keep degree ≤ 3	W2.3, W2.5

ADDED FOR CLARITY S008·6 · S008·11 · ESA Jan-24 s1 1b · ESA Jun-24 1c

Important points and why

1. About $(0,0)$ the constant term is often 0 ($\log 1 = 0$, $\sin 0 = 0$), so the series starts at first degree. ESA ★★ T008 00:07:57
2. "Up to three terms" means three non-zero terms; the lecture still writes all second-order terms. ESA ★★ T008 00:07:38 · T008 00:10:55
3. Collect and simplify at the end: $2x + y - 2x^2 - 2xy - \frac{y^2}{2}$. ESA ★ T008 00:11:11
4. Check a two-variable answer with the one-variable series of the combination (E2.2, Figure G2.1). ESA ★★★ ADDED FOR CLARITY S008·6 · S008·11

Why does $\log(2x+y+1)$ give $(2x+y) - \frac{(2x+y)^2}{2} + \ldots$?
It is $\log(1+t)$ with $t = 2x+y$, and $\log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \ldots$; the two-variable Maclaurin series must agree with it.
ADDED FOR CLARITY S008·6

Exam trap. For $e^x \log(1+y)$: $f_{xy} = \frac{e^x}{1+y}$, so $f_{xy}(0,0) = 1$, not 0 as slide 008·12 lists.
ADDED FOR CLARITY S008·12

Exam trap. "First six terms" of $e^x \log(1+y)$ needs third order: $y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$. The slide stops at three. ADDED FOR CLARITY S008·12 · S008·13

Worked examples

W2.1 · $\log(2x+y+1)$ about the origin (slides 008·5–6) ESA ★★

GIVEN Expand $\log(2x+y+1)$ about $(0,0)$ up to three terms.

STEP 1 $f = \log 1 = 0$; $f_x = \frac{2}{2x+y+1} = 2$; $f_y = \frac{1}{2x+y+1} = 1$.

STEP 2 $f_{xx} = \frac{-4}{(2x+y+1)^2} = -4$; $f_{xy} = \frac{-2}{(\ldots)^2} = -2$; $f_{yy} = \frac{-1}{(\ldots)^2} = -1$.

STEP 3 E2.1: $2x + y + \frac{1}{2!}\{-4x^2 - 4xy - y^2\}$.

ANSWER
$$\log(2x+y+1) \approx 2x + y - 2x^2 - 2xy - \frac{y^2}{2} = (2x+y) - \frac{(2x+y)^2}{2}$$

CHECK E2.2 with $t = 2x+y$; sympy agrees. The lecture's spoken simplification "$-xy$" (T008 00:11:11) should be $-2xy$.

S008·5 · S008·6 · T008 00:06:34 · T008 00:11:11

W2.2 · e^{x-y} about the origin (slides 008·10–11) ESA ★★

GIVEN Find the Taylor expansion of e^{x-y} at $(0,0)$.

STEP 1 At $(0,0)$ every derivative is ± 1: $f_x = 1$, $f_y = -1$, $f_{xx} = 1$, $f_{xy} = -1$, $f_{yy} = 1$.

STEP 2 Third order: $f_{xxx} = 1$, $f_{xxy} = -1$, $f_{xyy} = 1$, $f_{yyy} = -1$.

STEP 3 E2.1 + E1.2: $1 + x - y + \frac{1}{2}(x^2 - 2xy + y^2) + \frac{1}{6}(x^3 - 3x^2y + 3xy^2 - y^3)$.

ANSWER $$\boxed{e^{x-y} \approx 1 + (x-y) + \frac{(x-y)^2}{2} + \frac{(x-y)^3}{6}}$$

CHECK The slide stops at second order; the third-degree term is added (sympy). E2.2 with $t = x - y$.

S008·10 · S008·11

W2.3 · $e^x \log(1+y)$, first six terms (slides 008·12–13) ESA ★★

GIVEN Find the first six terms of the expansion of $e^x \log(1+y)$ about $(0,0)$.

STEP 1 $f = 0$, $f_x = e^x \log(1+y) = 0$, $f_y = \frac{e^x}{1+y} = 1$.

STEP 2 $f_{xx} = 0$, $f_{xy} = \frac{e^x}{1+y} = 1$ (slide 008·12 prints 0), $f_{yy} = -\frac{e^x}{(1+y)^2} = -1$.

STEP 3 Third order: $f_{xxx} = 0$, $f_{xxy} = 1$, $f_{xyy} = -1$, $f_{yyy} = \frac{2e^x}{(1+y)^3} = 2$.

STEP 4 $y + \frac{1}{2}(2xy - y^2) + \frac{1}{6}(3x^2y - 3xy^2 + 2y^3)$.

ANSWER $$\boxed{e^x \log(1+y) \approx y + xy - \frac{y^2}{2} + \frac{x^2y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}}$$

CHECK The slide gives the first three ($y + xy - \frac{y^2}{2}$); the other three are added. Product of $1 + x + \frac{x^2}{2}$ and $y - \frac{y^2}{2} + \frac{y^3}{3}$; sympy agrees.

S008·12 · S008·13

W2.4 · log(1 + x − y) to third degree (ESA) ESA ★★★

GIVEN Expand $\log(1 + x - y)$ about the origin up to third-degree terms.

STEP 1 $f_x = 1$, $f_y = -1$; $f_{xx} = -1$, $f_{xy} = 1$, $f_{yy} = -1$; $f_{xxx} = 2$, $f_{xxy} = -2$, $f_{xyy} = 2$, $f_{yyy} = -2$.

STEP 2 $(x-y) + \frac{1}{2}(-x^2 + 2xy - y^2) + \frac{1}{6}(2x^3 - 6x^2y + 6xy^2 - 2y^3)$.

ANSWER $$\boxed{(x-y) - \frac{(x-y)^2}{2} + \frac{(x-y)^3}{3}}$$

CHECK E2.2 with $t = x - y$; sympy agrees.

ESA Jan-24 s1 1b

W2.5 · $e^{ax}\sin by$ to third degree (ESA) ESA ★★★

GIVEN Expand $e^{ax}\sin by$ about the origin up to third-degree terms.

STEP 1 $f = f_x = 0$, $f_y = b$; $f_{xx} = 0$, $f_{xy} = ab$, $f_{yy} = 0$.

STEP 2 $f_{xxx} = 0$, $f_{xxy} = a^2 b$, $f_{xyy} = 0$, $f_{yyy} = -b^3$.

STEP 3 $by + abxy + \frac{1}{6}(3a^2bx^2y - b^3y^3)$.

ANSWER $$\boxed{by + abxy + \frac{a^2 b\, x^2 y}{2} - \frac{b^3 y^3}{6}}$$

CHECK Product of the series of e^{ax} and $\sin by$; sympy agrees.

ESA Jan-24 s2 1b

W2.6 · sin(x + 2y) to third degree (ESA) ESA ★★★

GIVEN Expand $\sin(x+2y)$ up to third-degree terms about $(0,0)$.

STEP 1 $f_x = 1$, $f_y = 2$; all second partials 0.

STEP 2 $f_{xxx} = -1$, $f_{xxy} = -2$, $f_{xyy} = -4$, $f_{yyy} = -8$.

STEP 3 $x + 2y + \frac{1}{6}(-x^3 - 6x^2y - 12xy^2 - 8y^3)$.

ANSWER $$\boxed{(x+2y) - \frac{(x+2y)^3}{6}}$$

CHECK CW answer agrees; E2.2 with $t = x + 2y$.

ESA Jun-24 1c · CW U1 P25

Past-paper questions

ESA Jan 2024 · 7 marks · UE23MA141A set 1, Q1.b · solved in W2.4

Expand $\log(1 + x - y)$ about origin up to third degree terms.

ESA Jan 2024 · 7 marks · UE23MA141A set 2, Q1.b · solved in W2.5

Expand $e^{ax}\sin by$ about origin up to third degree terms.

ESA June 2024 · 6 marks · UE23MA141A Q1.c · solved in W2.6

Expand $f(x,y) = \sin(x+2y)$ in Taylor's series up to third degree terms about the point $(0,0)$.

SHORT-NOTE RECAP

About $(0,0)$: $f + \{xf_x + yf_y\} + \frac{1}{2!}\{\dots\} + \frac{1}{3!}\{\dots\}$. $\log(2x+y+1) = (2x+y) - \frac{(2x+y)^2}{2}$; $e^{x-y} = 1 + (x-y) + \frac{(x-y)^2}{2} + \frac{(x-y)^3}{6}$; $e^x \log(1+y) = y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$. Check one-combination functions with the one-variable series.

3 Approximate values from a Taylor polynomial

To estimate f at an awkward point, expand about a nearby point where everything is known, then substitute the small h and k. Class 008 estimates $\tan^{-1}(xy)$ at $(0.9, -1.2)$ from $(1, -1)$; the class-work sheet estimates $\tan^{-1}\frac{0.9}{1.1}$ from $(1, 1)$. More terms give a better value.

Key terms and definitions

Approximate value. The value of a truncated Taylor polynomial at a point near the expansion point. S008·7 · T008 00:11:47

Formulas

E3.1 · Approximation by a Taylor polynomial ESA ★★

$$f(a+h, b+k) \approx f(a,b) + hf_x + kf_y + \frac{1}{2!}\{h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}\}$$

gives	A numerical estimate.
applies	h, k small; (a, b) a point where f and its derivatives are easy.
note	Choose (a, b) with "nice" values: $(1, -1)$ for $(0.9, -1.2)$.

S008·7 · S008·9 · T008 00:12:10

Graphs and diagrams

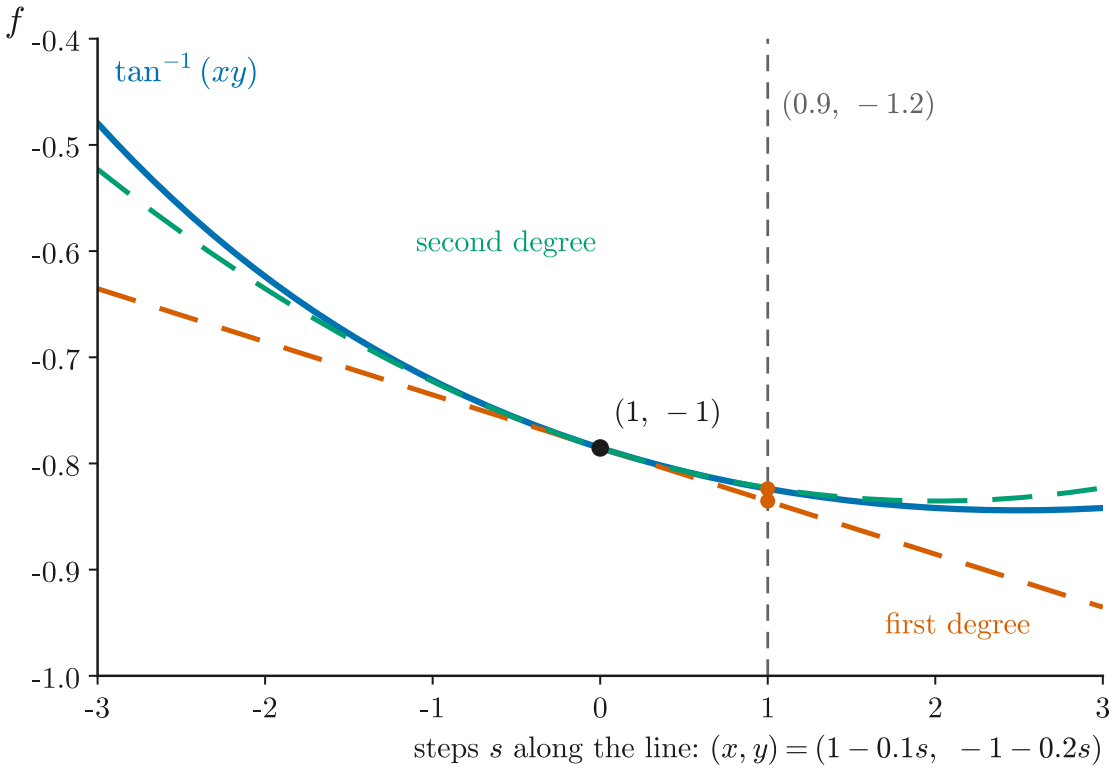
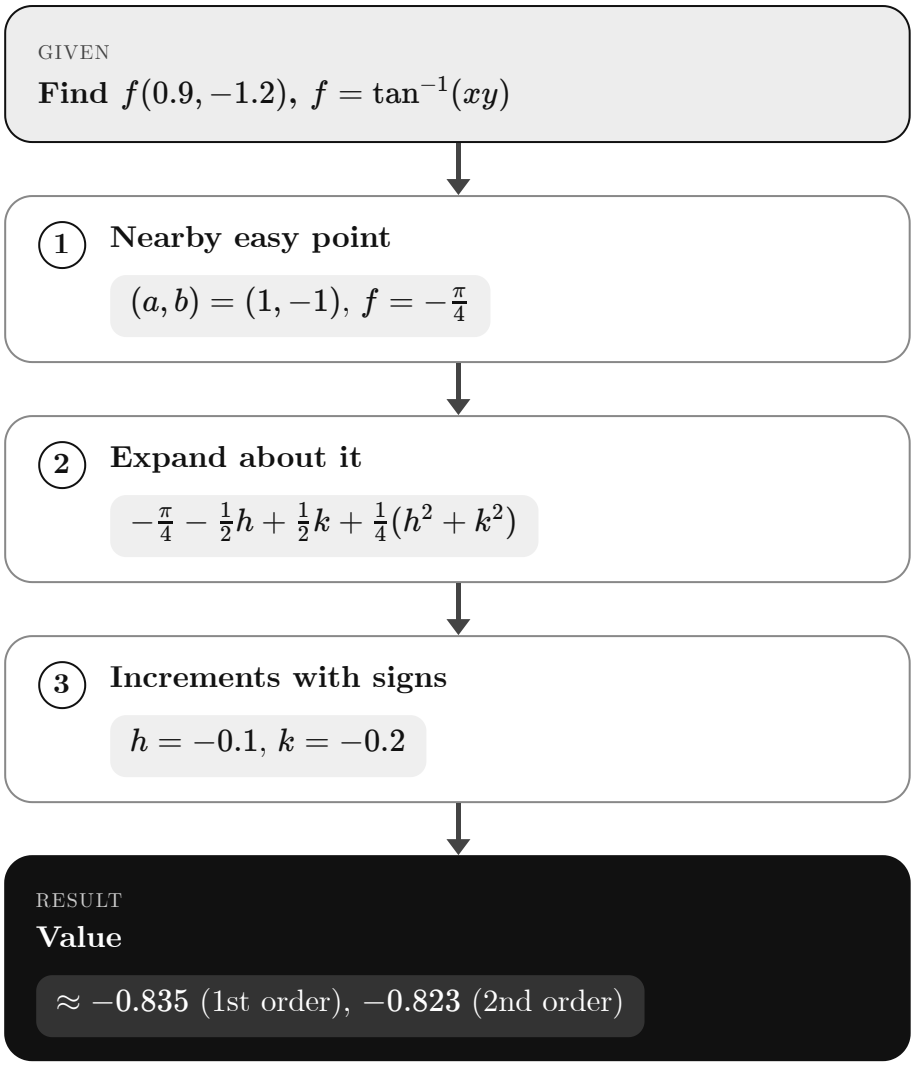


Figure G3.1. Approximating $\tan^{-1}(xy)$ at $(0.9, -1.2)$. ESA ★★

ADDED FOR CLARITY S008·7 · S008·9 · T008 00:14:18

How to read it. Values along the straight line from $(1, -1)$ (black dot, $s = 0$) through $(0.9, -1.2)$ ($s = 1$, grey dashed line). At $s = 1$ the first-degree polynomial (vermilion) gives -0.835, while the function (blue) is -0.824; the second degree (green) is almost on the curve there.

Flowcharts



Remark. Exact value -0.8238: the second-order term is worth keeping.

Figure F3.1. Approximate value from a Taylor polynomial ESA ★★

S008·7 · S008·9 · T008 00:12:10

Important points and why

1. Pick an expansion point near the target where f is easy: $(1, -1)$ for $(0.9, -1.2)$, $(1, 1)$ for $(1.1, 0.9)$. ESA ★★ T008 00:12:10 · CW U1 P26
2. "The more terms you take, the better the approximation": the first-order estimate of $\tan^{-1}(-1.08)$ is off by 0.012, the second-order one by 0.001. ESA ★★ T008 00:14:18
3. Substitute $h = x - a$ and $k = y - b$ with their signs: $x = 0.9$ gives $h = -0.1$, $y = -1.2$ gives $k = -0.2$. ESA ★★ T008 00:15:13 · T008 00:15:28

Why expand about (1, −1) to evaluate at (0.9, −1.2)?
At $(1,-1)$ every value is exact ($\tan^{-1}(-1)=-\frac{\pi}{4}$, simple derivatives), and $h=-0.1$, $k=-0.2$ are small, so the higher terms shrink quickly.

T008 00:12:10 · T008 00:12:48

Exam trap. With $x=0.9$, $h=x-1=-0.1$ (negative). A sign slip here changes the answer by 0.1. ADDED FOR CLARITY T008 00:15:13

Worked examples

W3.1 · tan⁻¹(xy) at (0.9, −1.2) (slides 008·7–9) — ESA ★★

GIVEN Find the Taylor series of $\tan^{-1}(xy)$ about $(1,-1)$ and hence approximate $f(0.9,-1.2)$. Let $h=x-1$, $k=y+1$.

STEP 1 $f=\tan^{-1}(-1)=-\frac{\pi}{4}$; $f_x=\frac{y}{1+x^2y^2}=-\frac{1}{2}$; $f_y=\frac{x}{1+x^2y^2}=\frac{1}{2}$.

STEP 2 $f_{xx}=\frac{-2xy^3}{(1+x^2y^2)^2}=\frac{1}{2}$; $f_{xy}=\frac{1-x^2y^2}{(1+x^2y^2)^2}=0$; $f_{yy}=\frac{-2x^3y}{(1+x^2y^2)^2}=\frac{1}{2}$.

STEP 3 Series: $-\frac{\pi}{4}-\frac{1}{2}h+\frac{1}{2}k+\frac{1}{4}(h^2+k^2)$.

STEP 4 $h=-0.1$, $k=-0.2$. First order: $-0.7854+0.05-0.1=-0.8354$. Adding $\frac{1}{4}(0.01+0.04)=0.0125$ gives -0.8229.

ANSWER $\boxed{\tan^{-1}(-1.08)\approx -0.835}$ (first order, as on the slide); ≈ -0.823 (second order)

CHECK Exact -0.82384 (sympy). The second-order value is ten times closer.

S008·7 · S008·8 · S008·9 · T008 00:11:53 · T008 00:15:42

W3.2 · tan⁻¹(0.9/1.1) (CW) — ESA ★★

GIVEN Using $\tan^{-1}\frac{y}{x}$ about $(1,1)$, compute $\tan^{-1}\frac{0.9}{1.1}$. Let $h=x-1$, $k=y-1$.

STEP 1 At $(1,1)$: $f=\frac{\pi}{4}$, $f_x=-\frac{1}{2}$, $f_y=\frac{1}{2}$, $f_{xx}=\frac{1}{2}$, $f_{xy}=0$, $f_{yy}=-\frac{1}{2}$.

STEP 2 $\frac{\pi}{4}-\frac{1}{2}h+\frac{1}{2}k+\frac{1}{4}h^2-\frac{1}{4}k^2$ with $h=0.1$, $k=-0.1$.

ANSWER $\boxed{\approx 0.6854}$

CHECK Exact 0.6857. The CW sheet prints 0.6904, an arithmetic slip.

CW U1 P26

SHORT-NOTE RECAP

Expand about a nearby easy point and substitute small h, k with signs. $\tan^{-1}(xy)$ at $(0.9,-1.2)$ from $(1,-1)$: -0.835 (first order), -0.823 (second); exact -0.824. $\tan^{-1}\frac{0.9}{1.1}\approx 0.6854$.

A Symbol glossary

Symbol	Name	Meaning	Appears in
(a, b)	Expansion point	$(1, \frac{\pi}{4})$, $(1, \frac{\pi}{2})$, $(1, 1)$, $(1, -1)$ or $(0, 0)$ in class 008	E1.1
$h,\ k$	Increments	$h = x - a$, $k = y - b$ (shorthand added for clarity)	E1.1, E3.1
$f_x, \ldots, f_{yy}$	Partial derivatives	Evaluated at the expansion point	E1.1, E2.1
$f_{xxx}, \ldots, f_{yyy}$	Third partials	Coefficients 1, 3, 3, 1	E1.2
t	Combination	$2x + y$, $x - y$, $x + 2y$ in the one-variable check	E2.2
s	Step along a line	Figure G3.1's parameter only	G3.1
$a,\ b$ (in $e^{ax} \sin by$)	Constants	Not the expansion point here (it is the origin)	W2.5

B Sources, coverage and gaps

Source	Path	Use
Deck 008 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.*	20 slides; content on 3–19
Transcript 008 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.*	16.2 min: slides 3–9
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 24–28 (with answers)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	6 in-scope Q1 parts
Video 008 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_*.mp4	Not opened: every garbled slide was settled from the transcript and by recomputing

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. A two-variable Taylor or Maclaurin expansion is asked in six of them, always a 6–7 mark Q1 part: Jul 2023 Q1.b and Dec 2023 Q1.b ($\sin x \sin y$ about $(\frac{\pi}{4}, \frac{\pi}{4})$, second order and third degree), May 2024 Q1.b ($e^x \cos y$ about $(1, \frac{\pi}{4})$, the same as slide 008·3), Jan 2024 set 1 Q1.b ($\log(1 + x - y)$), set 2 Q1.b ($e^{ax} \sin by$) and June 2024 Q1.c ($\sin(x + 2y)$). Four ask for third-degree terms. Jan 2024 sets 3 and 4 have no Taylor question. Approximate values (topic 3) were never asked in these papers; they appear in class and in CW problem 26.

Known gaps, conflicts and corrections

1. Slide 008·18 gives $f_{yy}(1, 1) = 3$ for $xy^2 + y\cos(x - y)$; it is 1. Slide 008·17's f_{yy} has the wrong sign on $y\cos(x - y)$: the correct $f_{yy} = 2x + 2\sin(x - y) - y\cos(x - y)$. Slide 008·19's final answer (with $f_{yy} = 1$) is correct.
2. Slide 008·12 lists $f_{xy}(0, 0) = 0$ for $e^x \log(1 + y)$; it is 1 ($f_{xy} = \frac{e^x}{1+y}$). Slide 008·13 uses the correct value.
3. Slide 008·12 asks for "the first six terms" of $e^x \log(1 + y)$ but slide 008·13 stops after three ($y + xy - \frac{y^2}{2}$). The six terms $y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3}$ are added for clarity (sympy).
4. Slide 008·11 computes the third-order derivatives of e^{x-y} but stops the series at second order; the third-degree term $\frac{(x-y)^3}{6}$ is added for clarity.
5. Slide 008·9 approximates $\tan^{-1}(0.9 \times (-1.2))$ with first-order terms only: $-\frac{\pi}{4} - 0.05 = -0.8354$. With the second-order terms of slide 008·8 it is -0.8229; the exact value is -0.8238 (sympy). Both are given.

6. Slides 008·6, 008·8 and 008·9 are partly garbled in the rendering; they were rebuilt from the transcript (T008 00:10:35, 00:14:35, 00:15:42) and checked by recomputing.
7. Slide 008·3 does not state the degree; the lecture says "up to second degree" (T008 00:00:21).
8. The lecture twice reads $f_{xx}(1, \frac{\pi}{4})$ and $f_{xy}(1, \frac{\pi}{4})$ as "e divided by two" (T008 00:03:47, 00:04:16); every value on slide 008·3 is $\pm\frac{e}{\sqrt{2}}$.
9. The lecture says "$f(0.9, -1.2)$ is approximately equal to $f(1, -1)$" (T008 00:12:10). That only motivates the choice of the expansion point; the approximation is the Taylor polynomial, not $f(1, -1) = -0.785$.
10. The lecture works slides 3–9 only (16.2 min). Slides 10–19 (e^{x-y}, $e^x \log(1+y)$, $\sin(xy)$, $xy^2 + y\cos(x-y)$) have no recorded explanation.
11. The class 007 theory is in its own pack; the formulas are repeated here without derivation.
12. Transcript recognition errors were corrected silently: "McLaren" / "my cleric series" → Maclaurin series, "e bar root" / "e pi root 2" → $e/\sqrt{2}$, "1 comma 5 by 4" → $(1, \frac{\pi}{4})$, "y minus 4" → $y - \frac{\pi}{4}$.

Coverage

Source	Citations	Slides cited (slide × citations)
S008	55	3×4, 4×1, 5×3, 6×6, 7×5, 8×1, 9×4, 10×4, 11×4, 12×4, 13×2, 14×4, 15×2, 16×3, 17×3, 18×3, 19×2
CW	6	
ESA	10	
T008	36	

Slides 008·1, 008·2 and 008·20 are title, session and closing slides. The lecture works slides 3–9 only; slides 10–19 have no recorded explanation and are cited from the slides alone.